2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

INTEGRATED SUSPENSION CONTROL MODULE (ISCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the integrated suspension control module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section.

For additional information, refer to: Vehicle Dynamic Suspension (204-05 Vehicle Dynamic Suspension, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B136B-64	Suspension Control Module Wake-up Signal - Signal plausibility failure	▪ Harness fault ▪ Central junction box not configured	▪ Check for related DTCs and refer to the relevant DTC index. Clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the integrated suspension control module power, ground, ignition and air suspension control signal circuits for open circuit. Repair circuit as required, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, confirm that the central junction box has the correct calibration data and software versions and install as required. Clear the DTCs and retest
C101D-11	Left Front Vertical Acceleration Sensor - Circuit short to ground	▪ Front left vertical acceleration sensor circuit - Short circuit to ground ▪ Vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front left vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101D-1C	Left Front Vertical Acceleration Sensor - Circuit	▪ Front left vertical acceleration sensor circuit - Short circuit	▪ Refer to the electrical circuit diagrams and check front left vertical acceleration sensor

	voltage out of range	to other circuit or short circuit to power, open circuit, high resistance ▪ Vertical acceleration sensor fault	circuit for short circuit to another circuit, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101D-22	Left Front Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Front left vertical acceleration sensor - Signal amplitude above maximum ▪ Front left vertical acceleration sensor signal circuit - Short circuit to another circuit ▪ Front left vertical acceleration sensor - Insecurely mounted ▪ Front left vertical acceleration sensor - Internal fault	▪ With vehicle parked on a level surface, read front left vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check front left vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101D-26	Left Front Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Front left vertical acceleration sensor - Signal not changing ▪ Front left vertical acceleration sensor signal circuit - Short circuit to another circuit ▪ Front left vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check front left vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101D-78	Left Front Vertical Acceleration Sensor -	▪ Front left vertical acceleration sensor - Alignment or	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the

	Alignment or adjustment incorrect	adjustment incorrect ▪ Front left vertical acceleration sensor - Bracket bent ▪ Front left vertical acceleration sensor damaged	DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101E-11	Right Front Vertical Acceleration Sensor - Circuit short to ground	▪ Front right vertical acceleration sensor circuit - Short circuit to ground ▪ Front right vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check front right vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101E-1C	Right Front Vertical Acceleration Sensor - Circuit voltage out of range	▪ Front right vertical acceleration sensor circuit - Short circuit to other circuit or short circuit to power, open circuit, high resistance ▪ Front right vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check front right vertical acceleration sensor circuit for short circuit to other circuit or short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101E-22	Right Front Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Front right vertical acceleration sensor - Signal amplitude above maximum ▪ Front right vertical acceleration sensor signal circuit - Short circuit to another circuit ▪ Front right vertical	▪ With vehicle parked on a level surface, read front right vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check front right vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as

		acceleration sensor - Insecurely mounted ▪ Front right vertical acceleration sensor - Internal fault	required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101E-26	Right Front Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Front right vertical acceleration sensor - Signal not changing ▪ Front right vertical acceleration sensor signal circuit - Short circuit to another circuit ▪ Front right vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check front right vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C101E-78	Right Front Vertical Acceleration Sensor - Alignment or adjustment incorrect	▪ Front right vertical acceleration sensor - Alignment or adjustment incorrect ▪ Front right vertical acceleration sensor - Bracket bent ▪ Front right vertical acceleration sensor damaged	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new vertical acceleration sensor. Clear the DTCs and retest
C1024-00	System Temporarily Disabled Due To Power Interruption During Driving - No sub type information	▪ Loss of power to control module whilst driving	▪ Refer to the electrical circuit diagrams and check power and ground circuits to the integrated suspension control module for intermittent or poor connections. Repair circuits as required, Clear the DTCs and retest

C102C-11	Right Rear Vertical Acceleration Sensor - Circuit short to ground	▪ Rear right vertical acceleration sensor circuit - Short circuit to ground ▪ Rear right vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check rear right vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If no wiring faults are present, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C102C-1C	Right Rear Vertical Acceleration Sensor - Circuit voltage out of range	▪ Rear right vertical acceleration sensor circuit - Partial short circuit to other circuit or short circuit to power, open circuit, high resistance ▪ Rear right vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check rear right vertical acceleration sensor circuit for partial short circuit to other circuit or short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If no wiring faults are present, check and install a new vertical acceleration sensor. Clear the DTCs and retest
C102C-22	Right Rear Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Rear right vertical acceleration sensor - Signal amplitude above maximum ▪ Rear right vertical acceleration sensor signal circuit - Short circuit to another circuit ▪ Rear right vertical acceleration sensor - Insecurely mounted ▪ Rear right vertical acceleration sensor - Internal fault	▪ With vehicle parked on a level surface, read rear right vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check rear right vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, install a new sensor. Clear the DTCs and retest

C102C-26	Right Rear Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Rear right vertical acceleration sensor - Signal not changing ▪ Rear right vertical acceleration sensor signal circuit - Short circuit to another circuit ▪ Rear right vertical acceleration sensor - Internal fault	▪ Refer to the electrical circuit diagrams and check rear right vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, install a new sensor. Clear the DTCs and retest
C102C-78	Right Rear Vertical Acceleration Sensor - Alignment or adjustment incorrect	▪ Rear right vertical acceleration sensor - Alignment or adjustment incorrect ▪ Rear right vertical acceleration sensor - Bracket bent ▪ Rear right vertical acceleration sensor damaged	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new sensor. Clear the DTCs and retest
C1032-64	Raise-Lower Switch - Signal plausibility failure	▪ Suspension raise and lower signals present simultaneously	▪ Check the operation of the suspension raise and lower switches. Clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the terrain response switchpack power and ground circuits for open circuit. Repair circuit as required, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network between the terrain response switchpack and the integrated suspension control module. Repair circuit as required, clear the DTCs and retest

			▪ If the fault persists, install a new terrain response switchpack. Clear the DTCs and retest
C1032-92	Raise-Lower Switch - Performance or incorrect operation	▪ Raise suspension switch jammed or obstructed ▪ Terrain response switchpack circuits - Short circuit to ground or short circuit to another switchpack circuit ▪ CAN fault ▪ Terrain response switchpack - Internal failure	▪ Check the operation of the suspension raise switch. Rectify as required. Clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the terrain response switchpack circuits for short circuit to ground or short circuit to another switchpack circuit. Repair circuit as required, Clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network between the terrain response switchpack and the integrated suspension control module. Repair circuit as required, Clear the DTCs and retest ▪ If the fault persists, install a new terrain response switchpack. Clear the DTCs and retest
C1032-94	Raise-Lower Switch - Unexpected operation	▪ Lower suspension switch jammed or obstructed ▪ Terrain response switchpack circuits - Short circuit to ground or short circuit to another switchpack circuit ▪ CAN fault ▪ Terrain response switchpack - Internal failure	▪ Check the operation of the suspension raise switch. Rectify as required. Clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the terrain response switchpack circuits for short circuit to ground or short circuit to another switchpack circuit. Repair circuit as required, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, complete a CAN network

			integrity test. Refer to the electrical circuit diagrams and check the CAN network between the terrain response switchpack and the integrated suspension control module. Repair circuit as required, Clear the DTCs and retest ▪ If the fault persists, install a new terrain response switchpack. Clear the DTCs and retest
C1056-66	Left Rear Air Spring Air Supply - Signal has too many transitions/events	▪ Leaking air spring or pipe to air spring ▪ Loose pipe fitting ▪ Rear cross-linking valve leak ▪ Corner valve leak to gallery ▪ Air suspension air supply unit fault - Relief pressure too low ▪ Vehicle has been driven while system in "Tight Tolerance" mode	▪ Check that there is pressure in the air spring. If the air spring is not under pressure, then use the manufacturer approved diagnostic system to open the corner valve and run the suspension air supply unit. Stop the suspension air supply unit if the gallery pressure exceeds 900 Kpa. Use leak detection spray on the connections to the air spring and the axle valve block to check for leaks while the air spring is being pressurised. If leak is detected in the air spring unit, replace the air spring damper assembly as required. If leak is detected in a connection, repair or replace the connector as required. Clear the DTCs and retest ▪ If no leak is evident, then inspect the air spring pipe and the associated axle valve block. If leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest ▪ To check for a leaking cross-link valve, with the vehicle around normal height, use the manufacturer approved diagnostic tool to lower one

corner on the axle being tested by at least 50mm. Leave the vehicle in this condition for at least 15 minutes. Observe height changes on the diagonally opposite corner of the other axle. The height of the diagonally opposite corner should increase and should remain in this position when the vehicle is left for 15 minutes. If the height of the diagonally opposite corner does not increase then the cross-link valve on the axle being tested is stuck wide open. If the height of the diagonally opposite corner does increase initially, but settles back to its starting position while the vehicle is left for 15 minutes then the cross linking valve is leaking. If a cross-link valve fault is detected, replace the faulty valve block as required. Clear the DTCs and retest

- To check for a leaking corner valve, use the manufacturer approved diagnostic tool to open the exhaust valve, then close all valves and monitor gallery pressure to see if it continues to increases after 2 minutes. If the gallery pressure continues to rise then it is likely that a corner valve or the reservoir is leaking. Disconnect the front gallery pipe from the rear valve block, fit a tight fitting hose over the end of this pipe and place the other end in a container of water. If a continuous flow of bubbles is observed then a front corner valve is leaking and the front valve block should be replaced. If no bubbles are observed then fit a temporary pipe to the front gallery port of the rear valve

				block and place the other end in a container of water. If a continuous flow of bubbles is observed then a rear corner valve or the reservoir valve may be leaking and the rear valve block should be replaced. Clear the DTC and retest ▪ Use the manufacturer approved diagnostic tool to vent the common gallery via the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer. Connect a calibrated pressure gauge tool directly into the delivery port. (Note that an 8mm connector will be required). Run the suspension air supply unit and check that the unit can generate a minimum of 18 bar air pressure. If 18 bar or greater can be generated, then the unit is operating correctly. Remove the pressure gauge and ensure that all pipe work is re-fitted correctly ▪ If pressure is lower than 18 bar, disconnect the suspension air supply unit intake pipe and run the suspension air supply unit again. If 18 bar pressure or greater can be generated with the intake pipe disconnected, then check for obstructions/blockages in the intake pipes and filter. Repair or replace intake system as required ▪ If a minimum of 18 bar pressure cannot be generated with the intake pipe disconnected, check and install a new air suspension air supply unit as required. Clear the DTCs and retest	

			▪ If the fault persists, using the manufacturer approved diagnostic tool, check the configuration status of the integrated suspension control module and reset the module to normal operation as required
C1058-66	Right Rear Air Spring Air Supply - Signal has too many transitions/events	▪ Leaking air spring or pipe to air spring ▪ Loose pipe fitting ▪ Rear cross-linking valve leak ▪ Corner valve leak to gallery ▪ Air suspension air supply unit fault - Relief pressure too low ▪ Vehicle has been driven while system in "Tight Tolerance" mode	▪ Check that there is pressure in the air spring. If the air spring is not under pressure, then use the manufacturer approved diagnostic system to open the corner valve and run the suspension air supply unit. Stop the suspension air supply unit if the gallery pressure exceeds 900 Kpa. Use leak detection spray on the connections to the air spring and the axle valve block to check for leaks while the air spring is being pressurised. If leak is detected in the air spring unit, replace the air spring damper assembly as required. If leak is detected in a connection, repair or replace the connector as required. Clear the DTCs and retest ▪ If no leak is evident, then inspect the air spring pipe and the associated axle valve block. If leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest ▪ To check for a leaking cross-link valve, with the vehicle around normal height, use the manufacturer approved diagnostic tool to lower one corner on the axle being tested by at least 50mm. Leave the vehicle in this condition for at least 15 minutes. Observe height changes on the diagonally

opposite corner of the other axle. The height of the diagonally opposite corner should increase and should remain in this position when the vehicle is left for 15 minutes. If the height of the diagonally opposite corner does not increase then the cross-link valve on the axle being tested is stuck wide open. If the height of the diagonally opposite corner does increase initially, but settles back to its starting position while the vehicle is left for 15 minutes then the cross linking valve is leaking. If a cross-link valve fault is detected, replace the faulty valve block as required. Clear the DTCs and retest

- To check for a leaking corner valve, use the manufacturer approved diagnostic tool to open the exhaust valve, then close all valves and monitor gallery pressure to see if it continues to increases after 2 minutes. If the gallery pressure continues to rise then it is likely that a corner valve or the reservoir is leaking. Disconnect the front gallery pipe from the rear valve block, fit a tight fitting hose over the end of this pipe and place the other end in a container of water. If a continuous flow of bubbles is observed then a front corner valve is leaking and the front valve block should be replaced. If no bubbles are observed then fit a temporary pipe to the front gallery port of the rear valve block and place the other end in a container of water. If a continuous flow of bubbles is observed then a rear corner valve or the reservoir valve may be leaking and the

				rear valve block should be replaced. Clear the DTCs and retest ▪ Use the manufacturer approved diagnostic tool to vent the common gallery via the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer. Connect a calibrated pressure gauge tool directly into the delivery port. (Note that an 8mm connector will be required). Run the suspension air supply unit and check that the unit can generate a minimum of 18 bar air pressure. If 18 bar or greater can be generated, then the unit is operating correctly. Remove the pressure gauge and ensure that all pipe work is re-fitted correctly ▪ If pressure is lower than 18 bar, disconnect the suspension air supply unit intake pipe and run the suspension air supply unit again. If 18 bar pressure or greater can be generated with the intake pipe disconnected, then check for obstructions/blockages in the intake pipes and filter. Repair or replace intake system as required ▪ If a minimum of 18 bar pressure cannot be generated with the intake pipe disconnected, check and install a new air suspension air supply unit as required. Clear the DTCs and retest ▪ If the fault persists, using the manufacturer approved diagnostic tool, check the configuration status of the integrated suspension control module and reset the	

			module to normal operation as required
C105A-01	Actuator Circuit Group A - General electrical failure	▪ Fuse failure ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module air suspension power supply fuse and circuit - Circuit reference VBATT AIR - For short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension valve circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C105A-19	Actuator Circuit Group A - Circuit current above threshold	▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module air suspension power supply circuit - Circuit reference VBATT AIR - For short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension valve circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105A-1D	Actuator Circuit Group A - Circuit current out of range	▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module air suspension power supply circuit - Circuit reference

			VBATT AIR - For short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension valve circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105A-64	Actuator Circuit Group A - Signal plausibility failure	▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module air suspension power supply circuit - Circuit reference VBATT AIR - For short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension valve circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-01	Actuator Circuit Group B - General electrical failure	▪ Fuse failure ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply fuse and circuit - Circuit reference VBATT CVD - For short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to

			ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-19	Actuator Circuit Group B - Circuit current above threshold	▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply circuit - Circuit reference VBATT CVD - For short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-1D	Actuator Circuit Group B - Circuit current out of range	▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply circuit - Circuit reference VBATT CVD - For short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-64	Actuator Circuit Group B - Signal	▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the

	plausibility failure		integrated suspension control module continuously variable damper power supply circuit - Circuit reference VBATT CVD - For short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C110C-01	Left Front Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Front left damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear DTC and retest ▪ If the fault persists, install a new shock absorber
C110C-12	Left Front Damper Solenoid - Circuit short to battery	▪ Front left damper solenoid circuit - Short circuit to power ▪ Front left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to power. Repair circuit as required, clear DTC and retest ▪ If the fault persists, install a new shock absorber
C110C-14	Left Front Damper Solenoid - Circuit short to ground or open	▪ Front left damper solenoid circuit - Short circuit to ground, open circuit, high resistance ▪ Front left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber

C110C-19	Left Front Damper Solenoid - Circuit current above threshold	▪ Front left damper solenoid circuit - Current above threshold	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest
C110C-1D	Left Front Damper Solenoid - Circuit current out of range	▪ Front left damper solenoid circuit - Current out of range ▪ Front left damper solenoid circuit - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110C-64	Left Front Damper Solenoid - Signal plausibility failure	▪ Front left damper solenoid - Signal plausibility failure ▪ Front left damper solenoid measured current control loop failed ▪ Front left damper solenoid circuit - Open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check front left damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C110D-01	Right Front Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Front right damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110D-	Right Front	▪ Front right damper	▪ Refer to the electrical circuit

12	Damper Solenoid - Circuit short to battery	solenoid circuit - Short circuit to power ▪ Front right damper solenoid failure	diagrams and check front right damper solenoid circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110D-14	Right Front Damper Solenoid - Circuit short to ground or open	▪ Front right damper solenoid circuit - Short circuit to ground, open circuit, high resistance ▪ Front right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110D-18	Right Front Damper Solenoid - Circuit current below threshold	▪ Front right damper solenoid circuit - Current below threshold ▪ Front right damper actuator - Open circuit at startup	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest
C110D-1D	Right Front Damper Solenoid - Circuit current out of range	▪ Front right damper solenoid circuit - Current out of range ▪ Front right damper solenoid circuit - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110D-64	Right Front Damper Solenoid - Signal plausibility failure	▪ Front right damper solenoid - Signal plausibility failure ▪ Front right damper solenoid measured current control loop failed	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest

		▪ Front right damper solenoid circuit - Open circuit, high resistance	▪ Check front right damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C110E-01	Left Rear Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Rear left damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110E-12	Left Rear Damper Solenoid - Circuit short to battery	▪ Rear left damper solenoid circuit - Short circuit to power ▪ Rear left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110E-14	Left Rear Damper Solenoid - Circuit short to ground or open	▪ Rear left damper solenoid circuit - Short circuit to ground, open circuit, high resistance ▪ Rear left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110E-19	Left Rear Damper Solenoid - Circuit current above threshold	▪ Rear left damper solenoid circuit - Current above threshold	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest
C110E-1D	Left Rear Damper Solenoid - Circuit current out of	▪ Rear left damper solenoid circuit - Current out of range	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for

	range	▪ Rear left damper solenoid circuit - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left damper solenoid failure	short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110E-64	Left Rear Damper Solenoid - Signal plausibility failure	▪ Rear left damper solenoid - Signal plausibility failure ▪ Rear left damper solenoid measured current control loop failed ▪ Rear left damper solenoid circuit - Open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check rear left damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C110F-01	Right Rear Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Rear right damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110F-12	Right Rear Damper Solenoid - Circuit short to battery	▪ Rear right damper solenoid circuit - Short circuit to power ▪ Rear right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to power. Repair circuit as required, clear DTC and retest ▪ If the fault persists, install a new shock absorber
C110F-14	Right Rear Damper Solenoid - Circuit short to ground or open	▪ Rear right damper solenoid circuit - Short circuit to ground, open circuit,	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to ground,

		high resistance ▪ Rear right damper solenoid failure	open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110F-19	Right Rear Damper Solenoid - Circuit current above threshold	▪ Rear right damper solenoid circuit - Current above threshold	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest
C110F-1D	Right Rear Damper Solenoid - Circuit current out of range	▪ Rear right damper solenoid circuit - Current out of range ▪ Rear right damper solenoid circuit - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new shock absorber
C110F-64	Right Rear Damper Solenoid - Signal plausibility failure	▪ Rear right damper solenoid - Signal plausibility failure ▪ Rear right damper solenoid measured current control loop failed ▪ Rear right damper solenoid circuit - Open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check rear right damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1111-22	Control Lateral Acceleration - Signal amplitude > maximum	▪ Integrated suspension control module - Internal failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, contact

			the Jaguar Land Rover technical helpdesk for further guidance
C1124-56	Height Sensor(s) - Invalid / incomplete configuration	**NOTE:** NOTE air suspension requires re-calibration - Integrated suspension control module has been swapped from another vehicle Height sensor bracket damaged/bent (after calibration) - Gateway module has been changed and not re-coded	- Using the manufacturer approved diagnostic system, reconfigure the relevant module(s) as required and calibrate the ride height. Clear the DTCs and retest
C112F-72	Air Spring Valve - Actuator stuck open	- Corner valve stuck open (fully or partially) - Vehicle driven while system in "Tight Tolerance" mode	- Complete corner valve checks. If necessary, clear tight tolerance mode. Clear the DTC and retest
C1130-53	Air Spring Air Supply - Deactivated	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to deflate	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to deflate - Ignore/clear this DTC

		▪ Diagnostic routine activity only	
C1130-66	Air Spring Air Supply - Signal has too many transitions/events	▪ Leaking air spring or pipe to air spring ▪ Loose pipe fitting ▪ Corner valve leak to gallery ▪ Air suspension air supply unit fault - Relief pressure too low ▪ Vehicle has been driven while system in "Tight Tolerance" mode	▪ Check that there is pressure in the air spring. If the air spring is not under pressure, then use the manufacturer approved diagnostic system to open the corner valve and run the suspension air supply unit. Stop the suspension air supply unit if the gallery pressure exceeds 900 Kpa. Use leak detection spray on the connections to the air spring and the axle valve block to check for leaks while the air spring is being pressurised. If leak is detected in the air spring unit, replace the air spring damper assembly as required. If leak is detected in a connection, repair or replace the connector as required. Clear the DTCs and retest ▪ If no leak is evident, then inspect the air spring pipe and the associated axle valve block. If leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest ▪ To check for a leaking corner valve, use the manufacturer approved diagnostic tool to open the exhaust valve, then close all valves and monitor gallery pressure to see if it continues to increases after 2 minutes. If the gallery pressure continues to rise then it is likely that a corner valve or the reservoir is leaking. Disconnect the front gallery pipe from the rear valve block, fit a tight fitting hose over the end of this pipe and place the other end in a container of water. If a

				continuous flow of bubbles is observed then a front corner valve is leaking and the front valve block should be replaced. If no bubbles are observed then fit a temporary pipe to the front gallery port of the rear valve block and place the other end in a container of water. If a continuous flow of bubbles is observed then a rear corner valve or the reservoir valve may be leaking and the rear valve block should be replaced. Clear the DTCs and retest ▪ Use the manufacturer approved diagnostic tool to vent the common gallery via the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer. Connect a calibrated pressure gauge tool directly into the delivery port. (Note that an 8mm connector will be required). Run the suspension air supply unit and check that the unit can generate a minimum of 18 bar air pressure. If 18 bar or greater can be generated, then the unit is operating correctly. Remove the pressure gauge and ensure that all pipe work is re-fitted correctly ▪ If pressure is lower than 18 bar, disconnect the suspension air supply unit intake pipe and run the suspension air supply unit again. If 18 bar pressure or greater can be generated with the intake pipe disconnected, then check for obstructions/blockages in the intake pipes and filter. Repair or replace intake system as required		

			- If a minimum of 18 bar pressure cannot be generated with the intake pipe disconnected, check and install a new air suspension air supply unit as required. Clear DTCs and retest - If fault persists, using the manufacturer approved diagnostic tool, check the configuration status of the integrated suspension control module and reset the module to normal operation as required
C1130-7A	Air Spring Air Supply - Fluid leak or seal failure	- Detached or burst air pipe - Leaking air spring or pipe to air spring - Loose or leaking air pipe connection	- Check that there is pressure in the air spring. If the air spring is not under pressure, then use the manufacturer approved diagnostic system to open the corner valve and run the suspension air supply unit. Stop the suspension air supply unit if the gallery pressure exceeds 900 Kpa. Use leak detection spray on the connections to the air spring and the axle valve block to check for leaks while the air spring is being pressurised. If leak is detected in the air spring unit, replace the air spring damper assembly as required. If leak is detected in a connection, repair or replace the connector as required. Clear DTCs and retest - If no leak is evident, then inspect the air spring pipes and the associated axle valve blocks. If leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest
C1131-	Air Supply -	Detached or burst	Use the manufacturer

	92	Performance or incorrect operation	air pipe ▪ Loose or leaking air pipe connection between reservoir valve block and suspension air supply unit or front axle valve block ▪ Insufficient pressure from air suspension air supply unit ▪ Faulty air pressure sensor	approved diagnostic system to run the suspension air supply unit and pressurise the gallery. Stop the suspension air supply unit when a pressure of 10 bar is reached. Use leak detection spray on the connections to the suspension air supply unit and associated valve blocks to check for leaks while the system is being pressurised. If leak is detected in a connection, repair or replace the connector as required. Clear DTCs and retest ▪ If gallery cannot be pressurised, then inspect the front gallery pipe and the pipes to between the suspension air supply unit and associated valve blocks. If leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest ▪ Use the manufacturer approved diagnostic tool to vent the common gallery via the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer. Connect a calibrated pressure gauge tool directly into the delivery port. (Note that an 8mm connector will be required). Run the suspension air supply unit and check that the unit can generate a minimum of 18 bar air pressure. If 18 bar or greater can be generated, then the unit is operating correctly. Remove the pressure gauge and ensure that all pipe work is re-fitted correctly ▪ If pressure is lower than 18 bar, disconnect the suspension air supply unit	

intake pipe and run the suspension air supply unit again. If 18 bar pressure or greater can be generated with the intake pipe disconnected, then check for obstructions/blockages in the intake pipes and filter. Repair or replace intake system as required

- If a minimum of 18 bar pressure cannot be generated with the intake pipe disconnected, check and install a new air suspension air supply unit as required. Clear the DTCs and retest
- To check air pressure sensor, use the manufacturer approved diagnostic tool to vent the common gallery via the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer and connect a T-piece fitting. Connect one port of the T-piece to the blue delivery port via a short length of 8mm pipe. Connect the other port of the T-piece to a calibrated pressure gauge tool (Note that an 8mm connector will be required). Using the manufacturer approved diagnostic tool, run the suspension air supply unit until the pressure is in the range 1500 - 1800 Kpa. Read the system pressure using the manufacturer approved diagnostic tool. Divide the measurement in Kpa by 100 then subtract 1 at sea level (0.9 at 900m elevation, 0.8 at 1900m elevation, or 0.7 at 3000m elevation) to convert to bar and compare the result with that displayed by the pressure gauge. If the readings are not within 2.0 bar of each other then the

			pressure sensor is likely to be faulty. In the case of a faulty pressure sensor, check and install a new valve block as required
C1138-92	Access Switch - Performance or incorrect operation	▪ Vehicle access switch (switch on driver's door) circuit - Short circuit to ground ▪ Driver's door switch pack fault	▪ Check the operation of the vehicle access switch and repair/renew as necessary ▪ Refer to the electrical circuit diagrams and check the vehicle access switch circuit from the driver's door to the integrated suspension control module for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new driver's door switch pack. Clear the DTCs and retest
C1A03-14	Left Front Height Sensor - Circuit short to ground or open	▪ Height sensor connector loose ▪ Front left height sensor circuit - Short circuit to ground, open circuit, high resistance ▪ Front left height sensor circuit - Short circuit to another circuit ▪ Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required

			▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest
C1A03-17	Left Front Height Sensor - Circuit voltage above threshold	▪ Front left height sensor circuit - Short circuit to another circuit ▪ Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required

			▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest
C1A03-21	Left Front Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front left height sensor circuit - Short circuit to ground ▪ Front left height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps

				below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that	

				the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure	
	C1A03-22	Left Front Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front left height sensor circuit - Short circuit to ground ▪ Front left height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control	

module and follow steps below

- Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required
- Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only
- Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest
- To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the

			mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A03-26	Left Front Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Front left height sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check left front height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A03-27	Left Front Height Sensor - Signal rate of change above threshold	▪ Height sensor or linkage loose ▪ Height sensor fault ▪ Front left height	▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector

		sensor circuit - Short circuit to ground, short circuit to power or intermittent open circuit, high resistance	for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check left front height sensor circuit for short circuit to ground, short circuit to power, or intermittent open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A03-64	Left Front Height Sensor - Signal plausibility failure	▪ Front left height sensor performance not as expected - Lowering slower than expected ▪ Axle valve block pipes connected incorrectly ▪ Blocked/damaged or crushed air spring pipe ▪ Blocked/damaged or crushed gallery pipe ▪ Height sensor fault ▪ Height sensor incorrectly installed ▪ Suspension movement obstructed ▪ Reservoir valve stuck open (mechanically)	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system.

			Clear the DTCs and retest ▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest
C1A03-92	Left Front Height Sensor - Performance or incorrect operation	▪ Front left height sensor performance not as expected - Lifting slower than expected ▪ Axle valve block pipes connected incorrectly ▪ Blocked/damaged or crushed air spring pipe ▪ Blocked/damaged or crushed gallery pipe ▪ Height sensor fault ▪ Height sensor incorrectly installed ▪ Suspension movement obstructed ▪ Reservoir valve stuck open (mechanically)	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. clear the DTCs and retest ▪ Visually inspect the height sensor for security, damage

			and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest
C1A04-14	Right Front Height Sensor - Circuit short to ground or open	▪ Height sensor connector loose ▪ Front right height sensor circuit - Short circuit to ground, open circuit, high resistance ▪ Front right height sensor circuit - Short circuit to another circuit ▪ Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as

			required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest
C1A04-17	Right Front Height Sensor - Circuit voltage above threshold	▪ Front right height sensor circuit - Short circuit to another circuit ▪ Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as

			required, clear the DTCs and retest - If the fault persists, install a new height sensor. clear the DTCs and retest
C1A04-21	Right Front Height Sensor - Signal amplitude < minimum	- Height sensor linkage not connected - Height sensor or bracket loose - Height sensor bracket bent - Incorrect height calibration - Height sensor linkage toggled - Front right height sensor circuit - Short circuit to ground - Front right height sensor circuit - Short circuit to another circuit - Height sensor electrical fault - Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated - Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database - To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below - Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required - Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only - Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor

				ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened,	

			found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A04-22	Right Front Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front right height sensor circuit - Short circuit to ground ▪ Front right height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not

				connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting		

			bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A04-26	Right Front Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Front right height sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check front right height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A04-27	Right Front Height Sensor - Signal rate of change above threshold	▪ Height sensor or linkage loose ▪ Height sensor fault ▪ Front right height sensor circuit - Short circuit to ground, short circuit to power or intermittent open circuit, high resistance	▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check front right height sensor circuit for short circuit to ground, short circuit to power, or intermittent open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A04-64	Right Front Height Sensor - Signal plausibility	▪ Front right height sensor performance not as expected -	

<table>
<tr><td></td><td>failure</td><td>Lowering slower than expected
Axle valve block pipes connected incorrectly
Blocked/damaged or crushed air spring pipe
Blocked/damaged or crushed gallery pipe
Height sensor fault
Height sensor incorrectly installed
Suspension movement obstructed
Reservoir valve stuck open (mechanically)</td><td>NOTE:

If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated
Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest
Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest
Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest
Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest
Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest</td></tr>
<tr><td>C1A04-92</td><td>Right Front Height Sensor - Performance or incorrect</td><td>Front right height sensor performance not as expected - Lifting slower than</td><td></td></tr>
</table>

	operation	expected ▪ Axle valve block pipes connected incorrectly ▪ Blocked/damaged or crushed air spring pipe ▪ Blocked/damaged or crushed gallery pipe ▪ Height sensor fault ▪ Height sensor incorrectly installed ▪ Suspension movement obstructed ▪ Reservoir valve stuck open (mechanically)	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. clear the DTCs and retest ▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest
C1A05-14	Left Rear Height Sensor - Circuit short to ground or open	▪ Height sensor connector loose ▪ Rear left height	

			sensor circuit - Short circuit to ground, open circuit, high resistance - Rear left height sensor circuit - Short circuit to another circuit - Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated - To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below - Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required - Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only - Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest - If the fault persists, install a new height sensor. clear the DTCs and retest

C1A05-17	Left Rear Height Sensor - Circuit voltage above threshold	- Rear left height sensor circuit - Short circuit to another circuit - Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated - To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below - Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required - Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only - Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest - If the fault persists, install a new height sensor. Clear the DTCs and retest

	C1A05-21	Left Rear Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear left height sensor circuit - Short circuit to ground ▪ Rear left height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and	

				retest - To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure	

	C1A05-22	Left Rear Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear left height sensor circuit - Short circuit to ground ▪ Rear left height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and	

					retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure	

C1A05-26	Left Rear Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Rear left height sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check rear left height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A05-27	Left Rear Height Sensor - Signal rate of change above threshold	▪ Height sensor or linkage loose ▪ Height sensor fault ▪ Rear left height sensor circuit - Short circuit to ground, short circuit to power or intermittent open circuit, high resistance	▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear DTC and retest ▪ If fault persists, refer to the electrical circuit diagrams and check rear left height sensor circuit for short circuit to ground, short circuit to power, or intermittent open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A05-64	Left Rear Height Sensor - Signal plausibility failure	▪ Rear left height sensor performance not as expected - Lowering slower than expected ▪ Axle valve block pipes connected incorrectly ▪ Blocked/damaged or crushed air spring pipe	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated

			▪ Blocked/damaged or crushed gallery pipe ▪ Height sensor fault ▪ Height sensor incorrectly installed ▪ Suspension movement obstructed ▪ Reservoir valve stuck open (mechanically)	▪ Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. clear the DTCs and retest ▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest	
	C1A05-92	Left Rear Height Sensor - Performance or incorrect operation	▪ Rear left height sensor performance not as expected - Lifting slower than expected		

			- Axle valve block pipes connected incorrectly - Blocked/damaged or crushed air spring pipe - Blocked/damaged or crushed gallery pipe - Height sensor fault - Height sensor incorrectly installed - Suspension movement obstructed - Reservoir valve stuck open (mechanically)	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated - Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest - Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest - Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest - Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest - Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest
	C1A06-14	Right Rear Height Sensor - Circuit short to ground or open	- Height sensor connector loose - Rear right height	

			sensor circuit - Short circuit to ground, open circuit, high resistance ▪ Rear right height sensor circuit - Short circuit to another circuit ▪ Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest

C1A06-17	Right Rear Height Sensor - Circuit voltage above threshold	▪ Rear right height sensor circuit - Short circuit to another circuit ▪ Height sensor - Internal failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest

	C1A06-21	Right Rear Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear right height sensor circuit - Short circuit to ground ▪ Rear right height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and

				retest - To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure		

	C1A06-22	Right Rear Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear right height sensor circuit - Short circuit to ground ▪ Rear right height sensor circuit - Short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins & terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and	

				retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure	

C1A06-26	Right Rear Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Rear right height sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check rear right height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A06-27	Right Rear Height Sensor - Signal rate of change above threshold	▪ Height sensor or linkage loose ▪ Height sensor fault ▪ Rear right height sensor circuit - Short circuit to ground, short circuit to power or intermittent open circuit, high resistance	▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check rear right height sensor circuit for short circuit to ground, short circuit to power, or intermittent open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A06-64	Right Rear Height Sensor - Signal plausibility failure	▪ Rear right height sensor performance not as expected - Lowering slower than expected ▪ Axle valve block pipes connected incorrectly ▪ Blocked/damaged or crushed air spring pipe	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated

			▪ Blocked/damaged or crushed gallery pipe ▪ Height sensor fault ▪ Height sensor incorrectly installed ▪ Suspension movement obstructed ▪ Reservoir valve stuck open (mechanically)	▪ Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest ▪ Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest
	C1A06-92	Right Rear Height Sensor - Performance or incorrect operation	▪ Rear right height sensor performance not as expected - Lifting slower than expected ▪ Axle valve block pipes connected incorrectly ▪ Blocked/damaged or crushed air spring pipe ▪ Blocked/damaged or crushed gallery pipe ▪ Height sensor fault	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check the front and rear valve block pipes for correct routing and installation. Repair/renew as required,

			- Height sensor incorrectly installed - Suspension movement obstructed - Reservoir valve stuck open (mechanically)	clear the DTCs and retest - Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest - Refer to the approved diagnostic system for corner and reservoir checks. Repair/renew as required, clear the DTCs and retest - Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest - Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest
	C1A07-62	Cross Articulation - Signal compare failure	**NOTE:** This DTC may be set by a fault relating to any of the height sensors. It will only set if fault is present and vehicle speed is greater than 55kph (35mph) for more than 25 seconds Height sensor linkage	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated Visually inspect the height sensors, height sensor linkages and height sensor brackets for security, damage and correct orientation/installation. Check the sensor connectors for damage. Repair/renew as

		- damaged/bent (after calibration) - Height sensor bracket damaged/bent (after calibration) - Incorrect height calibration, e.g. height sensor removed and re-installed or renewed without re-calibrating - Height sensor circuit(s) - Short circuit to ground, short circuit to power, short to another circuit, open circuit, high resistance - Tire severely under-inflated or under-size tire fitted - Height sensor fault	- required. If any sensors, linkages or brackets require attention, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest - If fault persists, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest - If fault persists, refer to the electrical circuit diagrams and check height sensor circuits for short circuit to ground, short circuit to power, short to another circuit, open circuit, high resistance. Repair circuit(s) as required, clear the DTCs and retest - Check that correct tires are fitted and check tire pressures. Rectify as necessary. Clear the DTCs and retest - If fault persists, check that the vehicle is free of obstructions. Check the height sensor(s) for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height using the approved diagnostic system. Clear the DTCs and retest - If the fault persists, install new height sensor(s)
C1A08-11	Pressure Sensor Supply - Circuit short to ground	- Air suspension air pressure sensor circuit - Short circuit to ground - Air suspension air pressure sensor failure	- Refer to the electrical circuit diagrams and check air suspension air pressure sensor circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest - If the fault persists, install a new rear valve block

C1A08-17	Pressure Sensor Supply - Circuit voltage above threshold	▪ Air suspension air pressure sensor supply voltage above threshold ▪ Air suspension air pressure sensor circuit - Short circuit to power ▪ Air suspension air pressure sensor circuit - Short circuit to another circuit ▪ Air suspension air pressure sensor failure	▪ Refer to the electrical circuit diagrams and check air suspension air pressure sensor circuit for short circuit to power, short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new rear valve block
C1A08-1C	Pressure Sensor Supply - Circuit voltage out of range	▪ Air suspension air pressure sensor supply voltage out of range ▪ Air suspension air pressure sensor circuit - Short circuit to ground, short circuit to power, short circuit to another circuit ▪ Air suspension air pressure sensor failure	▪ Refer to the electrical circuit diagrams and check air suspension air pressure sensor circuit for short circuit to ground, short circuit to power, short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new rear valve block
C1A10-64	Pressure Fluctuates When System Inactive - Signal plausibility failure	▪ Signal plausibility failure ▪ Axle valve block pipes connected incorrectly ▪ Pneumatic parts disconnected while system active ▪ Corner valve stuck open (mechanically) ▪ Air suspension air pressure sensor circuit - Short circuit to ground, short circuit to power, short circuit to	▪ Check for an air spring leak. Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, check the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner, reservoir and exhaust valve checks. Check the corner

		another circuit, open circuit, high resistance ▪ Air suspension air pressure sensor fault (calibration drift) ▪ Pressure gallery leak ▪ Exhaust valve fault ▪ Air suspension air pressure sensor failure	valve for leaks. Repair/renew as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check air suspension air pressure sensor circuit for short circuit to ground, short circuit to power, short circuit to another circuit, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new rear valve block
C1A11-64	Pressure Increases When System Inactive - Signal plausibility failure	▪ Signal plausibility failure ▪ Axle valve block pipes connected incorrectly ▪ Corner valve internal leak ▪ Reservoir valve internal leak ▪ Air suspension air pressure sensor circuit - Short circuit to ground, short circuit to power, short circuit to another circuit, open circuit, high resistance ▪ Pressure sensor fault ▪ Exhaust valve fault	▪ Check for an air spring leak. Check the air harness for evidence of melting, crushing, kinking or collapsing. Repair/renew as required, clear the DTCs and retest ▪ If fault persists, check the reservoir pipes and the front and rear valve block pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner, reservoir and exhaust valve checks. Check the corner valve for leaks. Repair/renew as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check air suspension air pressure sensor circuit for short circuit to ground, short circuit to power, short circuit to another circuit, open circuit, high resistance. Repair circuit as required, clear DTC and retest ▪ If the fault persists, install a new rear valve block

C1A13-64	Pressure Does Not Decrease When Venting Gallery - Signal plausibility failure	▪ Signal plausibility failure ▪ Gallery pipe from rear valve block to suspension air supply unit blocked/damaged ▪ Air suspension exhaust silencer or exhaust pipe blocked/restricted ▪ Rear valve block pipes connected incorrectly ▪ Pressure sensor fault ▪ Exhaust valve stuck closed	▪ Visually check the gallery pipes for evidence of melting, crushing, kinking or collapsing. If pipes appear sound, disconnect the pipe at both ends and check for free air flow. Repair/renew pipes as required, clear the DTCs and retest ▪ Check the air suspension air supply unit exhaust silencer/exhaust pipe for blockages/restrictions. Repair as required, clear the DTCs and retest ▪ Starting with the vehicle around normal height, using the manufacturer approved diagnostic system, open the exhaust valve. If the rear suspension height lowers quickly as soon as the exhaust valve is opened then a rear air spring pipe and the front gallery pipe are fitted to the rear valve block incorrectly. Rectify as required, clear the DTCs and retest ▪ To check air pressure sensor, use the manufacturer approved diagnostic tool to vent the common gallery via the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer and connect a T-piece fitting. Connect one port of the T-piece to the blue delivery port via a short length of 8mm pipe. Connect the other port of the T-piece to a calibrated pressure gauge tool (Note that an 8mm connector will be required). Using the manufacturer approved diagnostic tool, run the suspension air supply unit until the pressure is in the range 1500 - 1800 Kpa. Read

				the system pressure using the manufacturer approved diagnostic tool. Divide the measurement in Kpa by 100 then subtract 1 at sea level (0.9 at 900m elevation, 0.8 at 1900m elevation, or 0.7 at 3000m elevation) to convert to bar and compare the result with that displayed by the pressure gauge. If the readings are not within 2.0 bar of each other then the pressure sensor is likely to be faulty. In the case of a faulty pressure sensor, check and install a new valve block as required. clear the DTCs and retest ▪ To check for a stuck exhaust valve, use the manufacturer approved diagnostic tool to run the suspension air supply unit until a gallery pressure of at least 500 Kpa is reached, then switch off the suspension air supply unit and ensure all valves are closed before opening the exhaust valve and noting any changes in gallery pressure. If the gallery pressure drops quickly to between 125 Kpa and 250 Kpa at sea level (115 Kpa and 240 Kpa at 900m elevation, 105 Kpa and 230 Kpa at 1900m elevation, or 95 Kpa and 220 Kpa at 3000m elevation) then the exhaust valve is operating correctly. If the gallery pressure cannot vent quickly below 220 Kpa to 250 Kpa (elevation dependent, see above), then there is a fault with the air suspension system or its exhaust valve. Disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer. Connect a calibrated pressure gauge tool directly into the delivery port. (Note	

			that an 8mm connector will be required). Run the suspension air supply unit to generate 3 to 4 bar and then actuate the exhaust valve. If the pressure cannot vent quickly to between 0.25 bar and 1.5 bar then the exhaust valve is the likely cause of failure. In this case, check and install a new suspension air supply unit as required. If the pressure can vent quickly to between 0.25 bar and 1.5 bar then check for a blockage in the pipe between the compressor and the rear valve block, and/or a blockage in the rear valve block, and/or a pressure sensor fault. In the case of a blocked pipe, repair or replace the faulty pipe as required. In the case of a blockage in the rear valve block or a pressure sensor fault, check and install a new rear valve block as required. clear the DTCs and retest
C1A18-64	Pressure Increase Too Rapid When Filling Reservoir - Signal plausibility failure	▪ Signal plausibility failure ▪ Reservoir valve stuck closed (mechanically) ▪ Reservoir pipe blocked/damaged ▪ Reservoir port blocked/restricted ▪ Pressure sensor fault	▪ Check the reservoir pipes for correct routing and installation. Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner, reservoir and exhaust valve checks. Check the corner valve for leaks. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, install a new rear valve block as required
C1A20-64	Pressure Increase Too Slow When Filling Reservoir - Signal plausibility failure	▪ Signal plausibility failure ▪ Gallery pipe air leak ▪ Reservoir pipe air	▪ Check the gallery pipes and reservoir pipes for leaks, correct routing and installation. Check the integrity of the air reservoir.

		leak ▪ Reservoir air leak ▪ Intake filter blocked/restricted ▪ Intake pipe blocked/restricted ▪ Air suspension air supply unit fault ▪ Corner valve stuck open	Repair/renew as required, clear the DTCs and retest ▪ Refer to the approved diagnostic system for corner, reservoir and exhaust valve checks. Check the corner valve for leaks. Repair/renew as required, clear the DTCs and retest ▪ Check the operation of the air suspension air supply unit. Repair or replace as required, clear the DTCs and retest ▪ Check the air suspension air supply unit intake filter, intake pipe and silencer for blockages/restrictions. Repair as required, clear the DTCs and retest
C1A24-64	No Temperature Increase When Compressor Requested - Signal plausibility failure	▪ Signal plausibility failure ▪ Air suspension air supply unit cylinder head temperature sensor circuit - Open circuit, high resistance ▪ Air suspension air supply unit cylinder head temperature sensor detached from cylinder head ▪ Air suspension air supply unit cylinder head temperature sensor fault ▪ Air suspension air supply unit fault	▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit cylinder head temperature sensor circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Check that the air suspension air supply unit cylinder head temperature sensor is securely attached to the cylinder head. Rectify as required, clear the DTCs and retest ▪ If the fault persists, install a new air suspension air supply unit cylinder head temperature sensor, clear the DTCs and retest ▪ If fault persists, check the operation of the air suspension air supply unit. Repair or replace as required, clear the DTCs and retest
C1A24-67	No Temperature Increase When	▪ Signal plausibility failure	▪ Use the manufacturer approved diagnostic tool to

	Compressor Requested - Signal incorrect after event	- Air suspension air supply unit cylinder head temperature sensor - Circuit fault - Air suspension air supply unit cylinder head temperature sensor not attached to the suspension air supply unit - Air suspension air supply unit cylinder head temperature sensor - Internal failure	check if the suspension air supply unit will operate. If the unit does not operate check the air suspension air supply unit fuses and relays and check the power and ground circuits for open circuit, high resistance. Repair circuit(s) as required, clear the DTCs and retest - If the unit will operate via the diagnostic tool, disconnect the control module connector and check the resistance between the terminals on the harness connector using a digital multimeter. If the ambient temperature is >0°C and the resistance exceeds 6 MegaOhms then the wiring and the thermistor should be investigated. Check the resistance between the relevant pins of the suspension air supply unit four-way connector. Repair circuit(s) as required, clear the DTCs and retest - If the fault persists, install a new air suspension air supply unit, clear the DTCs and retest
C1A27-11	Compressor Circuit - Circuit short to ground	- Air suspension air supply unit voltage present when suspension air supply unit not requested - Air suspension air supply unit isolation relay stuck closed - Air suspension air supply unit isolation relay circuit - Short circuit to ground - Air suspension air supply unit isolation relay - Short circuit	- Refer to the electrical circuit diagrams and check the air suspension air supply unit isolation relay circuits for short circuit to ground. Repair circuit(s) as required, clear the DTCs and retest - Refer to the electrical circuit diagrams and check the air suspension air supply unit isolation relay high current terminals for short circuit to one another. Repair circuit(s) as required, clear the DTCs and retest - If the fault persists, install a new air suspension air supply

		between high current terminals ▪ Air suspension air supply unit isolation relay - Internal failure	unit isolation relay. Clear the DTCs and retest
C1A27-12	Compressor Circuit - Circuit short to battery	▪ Air suspension air supply unit voltage present when suspension air supply unit not requested ▪ Air suspension air supply unit main relay stuck closed ▪ Air suspension air supply unit main relay circuit - Short circuit to ground ▪ Air suspension air supply unit main relay - Short circuit between high current terminals ▪ Air suspension air supply unit main relay - Internal failure ▪ Air suspension air supply unit isolation relay stuck closed ▪ Air suspension air supply unit isolation relay circuit - Short circuit to ground ▪ Air suspension air supply unit isolation relay - Short circuit between high current terminals ▪ Air suspension air supply unit isolation relay - Internal failure	▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit isolation relay circuits for short circuit to ground. Repair circuit(s) as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit isolation relay high current terminals for short circuit to one another. Repair circuit(s) as required, clear the DTCs and retest ▪ If the fault persists, install a new air suspension air supply unit isolation relay. Clear the DTCs and retest ▪ If fault persists, refer to the electrical circuit diagrams and check the air suspension air supply unit isolation relay circuits and terminals for short circuit to ground. Repair circuit(s) as required, clear the DTCs and retest ▪ If the fault persists, install a new air suspension air supply unit isolation relay. Clear the DTCs and retest
C1A27-13	Compressor Circuit - Circuit open	▪ Air suspension air supply unit motor disconnected	▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit motor power and ground

		- Feedback protection fuse fault - Air suspension air supply unit isolation relay stuck open	circuits for open circuit, high resistance. Repair circuit as required, clear DTC and retest - Refer to the electrical circuit diagrams and check the integrity of the feedback protection fuse and associated wiring in the rear fuse box. Repair or replace as required, clear the DTCs and retest - Refer to the electrical circuit diagrams and check the integrity of the air suspension air supply unit isolation relay and associated wiring in the rear fuse box. Repair or replace as required, clear the DTCs and retest
C1A27-14	Compressor Circuit - Circuit short to ground or open	- Integrated suspension control module supply fusible link in battery junction box failed/not installed - Air suspension relay/air suspension air supply unit supply fusible link in battery junction box failed/not installed - Air suspension air supply unit power supply circuit - Short circuit to ground, open circuit, high resistance - Air suspension relay fault - Integrated suspension control module - Internal failure	- Refer to the electrical circuit diagrams and check the integrity of the air suspension relay/air suspension air supply unit supply fusible link and the integrated suspension control module supply fusible link in the battery junction box. Repair or replace as required, clear the DTCs and retest - If fault persists, refer to the electrical circuit diagrams and check the air suspension air supply unit power supply circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest - If the fault persists, install a new air suspension relay. Clear the DTCs and retest - If the fault persists, install a new integrated suspension control module as required. Clear the DTCs and retest

C1A27-15	Compressor Circuit - Circuit short to battery or open	▪ Air suspension air supply unit main fuse fault ▪ Air suspension air supply unit main relay circuit - Open circuit, high resistance ▪ Air suspension air supply unit main relay stuck open ▪ Air suspension air supply unit motor supply circuit - Short circuit to ground	▪ Refer to the electrical circuit diagrams and check the integrity of the air suspension air supply unit main fuse and associated wiring in the rear fuse box. Repair or replace as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the integrity of the air suspension air supply unit main relay and associated wiring in the rear fuse box. Repair or replace as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit motor power supply circuits for short circuit to ground. Repair circuit as required, clear the DTCs and retest
C1A27-29	Compressor Circuit - Signal invalid	▪ Air suspension relay control voltage signal invalid ▪ Air suspension air supply unit main relay coil terminals - Short circuit to one another, open circuit, high resistance ▪ Air suspension air supply unit main relay circuits - Short circuit to power, short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the integrity of the air suspension air supply unit main relay and associated wiring in the rear fuse box. Repair or replace as required, clear the DTCs and retest
C1A31-11	Left Front Corner Valve - Circuit short to ground	▪ Front left corner valve circuit - Short circuit to ground ▪ Front left corner valve failure	▪ Refer to the electrical circuit diagrams and check the front left corner valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit

			diagrams and check the front left corner valve solenoid for short circuit to ground. If valve fault exists, check and install a new front valve block assembly as required. Clear the DTCs and retest
C1A31-12	Left Front Corner Valve - Circuit short to battery	▪ Front left corner valve circuit - Short circuit to power ▪ Front left corner valve failure	▪ Refer to the electrical circuit diagrams and check the front left corner valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the front left corner valve solenoid for short circuit to power. If valve fault exists, check and install a new front valve block assembly as required. Clear the DTCs and retest
C1A31-13	Left Front Corner Valve - Circuit open	▪ Front left corner valve circuit - Open circuit, high resistance ▪ Front valve block disconnected ▪ Front left corner valve failure	▪ Refer to the electrical circuit diagrams and check the front left corner valve circuit for open circuit, high resistance. Repair circuit as required, clear DTC and retest ▪ If fault persists, check integrity of front valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new front valve block assembly as required. Clear the DTCs and retest
C1A31-73	Left Front Corner Valve - Actuator stuck closed	▪ Front valve block disconnected ▪ Front left corner valve failure	▪ Clear DTC and retest. If fault persists, check integrity of front valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new front valve block assembly. Clear the DTCs and retest

C1A32-11	Right Front Corner Valve - Circuit short to ground	▪ Front right corner valve circuit - Short circuit to ground ▪ Front right corner valve failure	▪ Refer to the electrical circuit diagrams and check the front right corner valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the front right corner valve solenoid for short circuit to ground. If valve fault exists, check and install a new front valve block assembly. Clear the DTCs and retest
C1A32-12	Right Front Corner Valve - Circuit short to battery	▪ Front right corner valve circuit - Short circuit to power ▪ Front right corner valve failure	▪ Refer to the electrical circuit diagrams and check the front right corner valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the front right corner valve solenoid for short circuit to power. If valve fault exists, check and install a new front valve block assembly as required. Clear the DTCs and retest
C1A32-13	Right Front Corner Valve - Circuit open	▪ Front right corner valve circuit - Open circuit, high resistance ▪ Front valve block disconnected ▪ Front right corner valve failure	▪ Refer to the electrical circuit diagrams and check the front right corner valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of front valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new front valve block assembly. Clear the DTCs and retest
C1A32-	Right Front	▪ Front valve block	▪ Clear the DTCs and retest. If

73	Corner Valve - Actuator stuck closed	disconnected ▪ Front right corner valve failure	fault persists, check integrity of front valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new front valve block assembly. Clear the DTCs and retest
C1A33-11	Left Rear Corner Valve - Circuit short to ground	▪ Rear left corner valve circuit - Short circuit to ground ▪ Rear left corner valve failure	▪ Refer to the electrical circuit diagrams and check the rear left corner valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the rear left corner valve solenoid for short circuit to ground. If valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A33-12	Left Rear Corner Valve - Circuit short to battery	▪ Rear left corner valve circuit - Short circuit to power ▪ Rear left corner valve failure	▪ Refer to the electrical circuit diagrams and check the rear left corner valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the rear left corner valve solenoid for short circuit to power. If valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A33-13	Left Rear Corner Valve - Circuit open	▪ Rear left corner valve circuit - Open circuit, high resistance ▪ Rear valve block disconnected ▪ Rear left corner valve failure	▪ Refer to the electrical circuit diagrams and check the rear left corner valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of rear valve block

			connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new rear valve block assembly. Clear the DTCs and retest
C1A33-72	Left Rear Corner Valve - Actuator stuck open	▪ Corner valve stuck open (fully or partially) ▪ Vehicle driven while system in "Tight Tolerance" mode	▪ Complete corner valve checks. If necessary, clear tight tolerance mode. Clear the DTCs and retest
C1A33-73	Left Rear Corner Valve - Actuator stuck closed	▪ Corner valve stuck closed (fully or partially) ▪ Vehicle driven while system in "Tight Tolerance" mode	▪ Complete corner valve checks. If necessary, clear tight tolerance mode. Clear the DTCs and retest
C1A34-11	Right Rear Corner Valve - Circuit short to ground	▪ Rear right corner valve circuit - Short circuit to ground ▪ Rear right corner valve failure	▪ Refer to the electrical circuit diagrams and check the rear right corner valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the rear right corner valve solenoid for short circuit to ground. If valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A34-12	Right Rear Corner Valve - Circuit short to battery	▪ Rear right corner valve circuit - Short circuit to power ▪ Rear right corner valve failure	▪ Refer to the electrical circuit diagrams and check the rear right corner valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the rear right corner valve solenoid for short circuit to power. If

			valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A34-13	Right Rear Corner Valve - Circuit open	▪ Rear right corner valve circuit - Open circuit, high resistance ▪ Rear valve block disconnected ▪ Rear right corner valve failure	▪ Refer to the electrical circuit diagrams and check the rear right corner valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of rear valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new rear valve block assembly. Clear the DTCs and retest
C1A34-72	Right Rear Corner Valve - Actuator stuck open	▪ Corner valve stuck open (fully or partially) ▪ Vehicle driven while system in "Tight Tolerance" mode	▪ Complete corner valve checks. If necessary, clear tight tolerance mode. Clear the DTCs and retest
C1A34-73	Right Rear Corner Valve - Actuator stuck closed	▪ Corner valve stuck closed (fully or partially) ▪ Vehicle driven while system in "Tight Tolerance" mode	▪ Complete corner valve checks. If necessary, clear tight tolerance mode. Clear the DTCs and retest
C1A35-11	Reservoir Valve - Circuit short to ground	▪ Reservoir valve circuit - Short circuit to ground ▪ Reservoir valve failure	▪ Refer to the electrical circuit diagrams and check the reservoir valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the reservoir valve solenoid for short circuit to ground. If valve fault exists, check and install a new rear valve block

			assembly as required. Clear the DTCs and retest
C1A35-12	Reservoir Valve - Circuit short to battery	▪ Reservoir valve circuit - Short circuit to power ▪ Reservoir valve failure	▪ Refer to the electrical circuit diagrams and check the reservoir valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the reservoir valve solenoid for short circuit to power. If valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A35-13	Reservoir Valve - Circuit open	▪ Reservoir valve circuit - Open circuit, high resistance ▪ Rear valve block disconnected ▪ Reservoir valve failure	▪ Refer to the electrical circuit diagrams and check the reservoir valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of rear valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new rear valve block assembly. Clear the DTCs and retest
C1A36-11	Exhaust Valve - Circuit short to ground	▪ Exhaust valve circuit - Short circuit to ground ▪ Exhaust valve failure	▪ Refer to the electrical circuit diagrams and check the exhaust valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the exhaust valve solenoid for short circuit to ground. If valve fault exists, check and install a new suspension air supply unit as required. Clear the DTCs and retest

C1A36-12	Exhaust Valve - Circuit short to battery	▪ Exhaust valve circuit - Short circuit to power ▪ Exhaust valve failure	▪ Refer to the electrical circuit diagrams and check the exhaust valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the exhaust valve solenoid for short circuit to power. If valve fault exists, check and install a new suspension air supply unit as required. Clear the DTCs and retest
C1A36-13	Exhaust Valve - Circuit open	▪ Exhaust valve circuit - Open circuit, high resistance ▪ Exhaust valve disconnected ▪ Exhaust valve failure	▪ Refer to the electrical circuit diagrams and check the exhaust valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of rear valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new suspension air supply unit. Clear the DTCs and retest
C1A37-11	Front Cross-Link Valve - Circuit short to ground	▪ Front cross-link valve circuit - Short circuit to ground ▪ Front cross-link valve failure	▪ Refer to the electrical circuit diagrams and check the front cross-link valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the front cross-link valve solenoid for short circuit to ground. If valve fault exists, check and install a new front valve block assembly as required. Clear the DTCs and retest
C1A37-	Front Cross-Link	▪ Front cross-link valve	▪ Refer to the electrical circuit

12	Valve - Circuit short to battery	circuit - Short circuit to power ▪ Front cross-link valve failure	diagrams and check the front cross-link valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the front cross-link valve solenoid for short circuit to power. If valve fault exists, check and install a new front valve block assembly as required. Clear the DTCs and retest
C1A37-13	Front Cross-Link Valve - Circuit open	▪ Front cross-link valve circuit - Open circuit, high resistance ▪ Front valve block disconnected ▪ Front cross-link valve failure	▪ Refer to the electrical circuit diagrams and check the front cross-link valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of front valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new front valve block assembly. Clear the DTCs and retest
C1A38-11	Rear Cross-Link Valve - Circuit short to ground	▪ Rear cross-link valve circuit - Short circuit to ground ▪ Rear cross-link valve failure	▪ Refer to the electrical circuit diagrams and check the rear cross-link valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the rear cross-link valve solenoid for short circuit to ground. If valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A38-12	Rear Cross-Link Valve - Circuit short to battery	▪ Rear cross-link valve circuit - Short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear cross-link valve circuit for

		▪ Rear cross-link valve failure	short circuit to power. Repair circuit as required, Clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the rear cross-link valve solenoid for short circuit to power. If valve fault exists, check and install a new rear valve block assembly as required. Clear the DTCs and retest
C1A38-13	Rear Cross-Link Valve - Circuit open	▪ Rear cross-link valve circuit - Open circuit, high resistance ▪ Rear valve block disconnected ▪ Rear cross-link valve failure	▪ Refer to the electrical circuit diagrams and check the rear cross-link valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If fault persists, check integrity of rear valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new rear valve block assembly. Clear the DTCs and retest
C1A68-1C	Left Front Height Sensor Supply - Circuit voltage out of range	▪ Front left height sensor supply circuit - Voltage out of range ▪ Height sensor wiring harness - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system ▪ Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high

				resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest
	C1A69-1C	Right Front Height Sensor Supply - Circuit voltage out of range	▪ Front right height sensor supply circuit - Voltage out of range ▪ Height sensor wiring harness - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system ▪ Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new height sensor. Clear the DTCs and retest
	C1A70-1C	Left Rear Height Sensor Supply - Circuit voltage out of range	▪ Rear left height sensor supply circuit - Voltage out of range ▪ Height sensor wiring harness - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system

			- Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, Clear the DTCs and retest - If the fault persists, install a new height sensor. Clear the DTCs and retest
C1A71-1C	Right Rear Height Sensor Supply - Circuit voltage out of range	- Rear right height sensor supply circuit - Voltage out of range - Height sensor wiring harness - Short circuit to ground, short circuit to power, open circuit, high resistance - Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated - Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system - Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest - If the fault persists, install a new height sensor. Clear the DTCs and retest
C1A99-1C	Pressure Sensor - Circuit voltage out of range	- Air pressure sensor circuit - Voltage out of range - Air pressure sensor disconnected	- Refer to the electrical circuit diagrams and check air pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit

		▪ Air pressure sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air pressure sensor fault	as required, Clear the DTCs and retest ▪ If the fault persists, install a new rear valve block. Clear the DTCs and retest
C1B15-11	Sensor Supply Voltage B - Circuit short to ground	▪ Vertical accelerometer supply circuit(s) - Short circuit to ground ▪ Vertical accelerometer failure	▪ Refer to the electrical circuit diagrams and check the vertical accelerometer supply circuit(s) for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install new vertical accelerometer(s) as required. After installation, calibrate the accelerometer/integrated suspension control module, as required. Clear the DTCs and retest ▪ If fault persists, check control module sensor supply output voltage. Measured voltage should be between 4.75 volts and 4.96 volts. If output voltage is out of range, check and install a new integrated suspension control module as required. Clear the DTCs and retest
C1B15-17	Sensor Supply Voltage B - Circuit voltage above threshold	▪ Vertical accelerometer supply circuit(s) - Short circuit to power, short circuit to another circuit ▪ Vertical accelerometer failure	▪ Refer to the electrical circuit diagrams and check the vertical accelerometer supply circuit(s) for short circuit to power, short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install new vertical accelerometer(s) as required. After installation, calibrate the accelerometer/integrated suspension control module, as required. Clear the DTCs

			and retest ▪ If fault persists, check control module sensor supply output voltage. Measured voltage should be between 4.75 volts and 4.96 volts. If output voltage is out of range, check and install a new integrated suspension control module as required. Clear the DTCs and retest
C1B15-1C	Sensor Supply Voltage B - Circuit voltage out of range	▪ Vertical accelerometer supply circuit(s) - Short circuit to ground, short circuit to another circuit ▪ Vertical accelerometer failure	▪ Refer to the electrical circuit diagrams and check the vertical accelerometer supply circuit(s) for short circuit to ground, short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install new vertical accelerometer(s) as required. After installation, calibrate the accelerometer/integrated suspension control module, as required. Clear the DTCs and retest ▪ If fault persists, check control module sensor supply output voltage. Measured voltage should be between 4.75 volts and 4.96 volts. If output voltage is out of range, check and install a new integrated suspension control module as required. Clear the DTCs and retest
C1B18-2F	Module Power Supplies - Signal erratic	▪ Loss of power to the control module whilst driving	▪ Check the module connector(s) for security and integrity. Rectify as required ▪ Refer to the electrical circuit diagrams and check the power supply circuit(s) to the module for poor or intermittent connection. Repair circuit(s) as required, clear the DTCs and retest

C1B18-62	Module Power Supplies - Signal compare failure	▪ Signal compare failure ▪ Inconsistent battery and ignition voltages received by integrated suspension control module ▪ Integrated suspension control module supply circuit(s) - Short circuit to ground ▪ Integrated suspension control module supply circuit(s) - High resistance	▪ Check the module connector(s) for security and integrity. Rectify as required ▪ Refer to the electrical circuit diagrams and check the power supply circuit(s) to the module for short circuit to ground, high resistance. Repair circuit(s) as required, clear the DTCs and retest
C1B18-94	Module Power Supplies - Unexpected operation	▪ Integrated suspension control module cannot power down as required ▪ Integrated suspension control module - Internal failure	▪ Clear the DTCs and retest. If the fault persists, install a new integrated suspension control module
C1B21-1C	Compressor Brush Card Temperature Sensor - Circuit voltage out of range	▪ Air suspension air supply unit brush card temperature sensor circuit - Voltage out of range ▪ Air suspension air supply unit brush card temperature sensor circuit - Short circuit to ground, short circuit to power, high resistance ▪ Air suspension air supply unit brush card temperature sensor failure	▪ Refer to the electrical circuit diagrams and check the suspension air supply unit sensor temperature circuit between the self-leveling suspension air supply unit and the integrated suspension control module for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, Clear the DTCs and retest ▪ If the fault persists, install a new self-leveling suspension air supply unit. Clear the DTCs and retest

P1905-00	Control Module Configured for End-of-Line Test Mode - No sub type information	▪ Integrated suspension control module - Not correctly configured	▪ Using the manufacturer approved diagnostic system, configure the module as a new module. Clear the DTCs and retest
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (chassis or powertrain)	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTC and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Missing message from another control module via the high speed CAN bus (chassis or powertrain)	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the relevant control module (i.e. as indicated by the snapshot data) power and ground

			circuits for open circuit. Repair circuits as required, Clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN network (chassis and powertrain) between the relevant control module (i.e. as indicated by the snapshot data) and the integrated suspension control module
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (chassis/powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the high speed CAN bus (chassis/powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ Car configuration signal not received ▪ Car configuration file incorrect ▪ The integrated suspension control module is not compatible with the vehicle	▪ Using the manufacturer approved diagnostic system check/amend the car configuration file. Clear the DTCs and retest ▪ If fault persists, check the integrated suspension control module part number, install the correct part as required. Clear the DTCs and retest
U0300-57	Internal Control Module Software Incompatibility - Invalid/incomplete software component	▪ Incorrect software for the integrated suspension control module has been installed	▪ Using the manufacturer approved diagnostic system check/amend the module software. Clear the DTCs and retest
U1A14-00	CAN Initialization Failure - No sub type information	▪ CAN network harness short circuit or disconnected	▪ Refer to the electrical circuit diagrams and check the power and ground

			connections to the module ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network ▪ Refer to the electrical circuit diagrams and check the CAN network between the central junction box and the integrated suspension control module
U201A-54	Control Module Main Calibration Data - Missing calibration	▪ A new integrated suspension control module has been installed and is not correctly configured	▪ Using the manufacturer approved diagnostic system, configure the module as a new module. Clear the DTCs and retest
U201A-57	Control Module Main Calibration Data - Invalid/incomplete software component	▪ A new integrated suspension control module has been installed and is not correctly configured ▪ Incorrect calibration data has been installed ▪ Incorrect software has been installed	▪ Using the manufacturer approved diagnostic system, confirm correct calibration data and software versions and install as required. Clear the DTCs and retest
U2100-00	Initial Configuration Not Complete - No sub type information	▪ Car configuration data has not been loaded correctly ▪ New central junction box has been fitted but not initialized ▪ Central junction box fault	▪ Check/amend the car configuration file using the manufacturer approved diagnostic system. Clear the DTCs and retest ▪ Check the central junction box for related DTCs and refer to the relevant DTC index
U2101-00	Control Module Configuration Incompatible - No sub type information	▪ Car configuration data transmitted over CAN does not match integrated suspension control module internal	▪ Carry out the new module software installation procedure using the manufacturers approved diagnostic system. Clear the DTCs and retest

		configuration	
U210A-16	Temperature Sensor - Circuit voltage below threshold	▪ Air suspension air supply unit cylinder head temperature sensor circuit - Circuit voltage below threshold ▪ Air suspension air supply unit cylinder head temperature sensor circuit - Short circuit to ground or short circuit to another circuit	▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit cylinder head temperature sensor circuit between the suspension air supply unit and the integrated suspension control module for short circuit to ground or short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, install a new air suspension air supply unit
U2300-54	Central Configuration - Missing calibration	▪ Car configuration file mismatch with vehicle specification ▪ Integrated suspension control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, re-configure the integrated suspension control module with the latest level software
U2300-55	Central Configuration - Not configured	▪ Integrated suspension control module is not configured correctly ▪ Car configuration file mismatch with vehicle specification	

			NOTE: After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, re-configure the integrated suspension control module with the latest level software ▪ Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid / incomplete configuration	▪ Car configuration file mismatch with vehicle specification ▪ Integrated suspension control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, re-configure the integrated suspension control module with the latest level software
U2300-64	Central Configuration - Signal plausibility failure	▪ Integrated suspension control module is not configured correctly ▪ Car configuration file	

			mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, re-configure the integrated suspension control module with the latest level software ▪ Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary. Clear the DTCs and retest
	U3000-04	Control Module - System internal failures	▪ Control module - Internal failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, contact the Jaguar Land Rover technical helpdesk for further guidance
	U3000-05	Control Module - System programming failures	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to operate in tight tolerance mode	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to operate in tight tolerance mode ▪ This DTC is not a fault but is logged when the air suspension has been set in tight tolerance mode. Use

		▪ System programming failures	the manufacturers approved diagnostic equipment to set system into normal (customer) mode
U3000-43	Control Module - Special memory failure	▪ Control module - Internal failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new integrated suspension control module
U3000-45	Control Module - Program memory failure	▪ Control module - Internal failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new integrated suspension control module
U3000-47	Control Module - Watchdog/safety microcontroller failure	▪ Control module - Internal watchdog/safety microcontroller failure ▪ The control module has carried out a software reset	▪ Clear the DTCs and retest. If the fault persists, contact dealer technical support
U3000-54	Control Module - Missing calibration	▪ The control module has been replaced and has not been calibrated	▪ Using the manufacturer approved diagnostic system, calibrate the vehicle ride height. Clear the DTCs and retest
U3002-62	Vehicle Identification Number - Signal compare failure		▪ Using the manufacturer approved diagnostic system, reconfigure the relevant module(s) and calibrate the ride height. Clear the DTCs and retest

			NOTE: Air suspension requires re-calibration ▪ Signal compare failure ▪ Integrated suspension control module has been swapped from another vehicle ▪ Instrument cluster has been changed and not re-coded		
	U3003-1C	Battery Voltage - Circuit voltage out of range	▪ Power supply circuit - Voltage out of range (supply voltage at control module below 10.5V or above 18V for 30 seconds)	▪ Check the battery is in good condition and fully charged, refer to the battery care manual ▪ Refer to the starting and charging section of the workshop manual and check the performance of the charging system ▪ Refer to the electrical circuit diagrams and check power and ground circuit to the control module for faults, including intermittent high resistance. Repair circuit, clear the DTCs and retest	
	U3003-62	Battery Voltage - Signal compare failure	▪ Signal compare failure ▪ High resistance connections	▪ Check the battery is in good condition and fully charged, refer to the battery care manual ▪ Refer to the starting and charging section of the workshop manual and check the performance of the charging system ▪ Refer to the electrical circuit diagrams and check power and ground circuit to the	

			control module for faults, including intermittent high resistance. Repair circuit, clear the DTCs and retest
U300A-1C	Ignition Switch - Circuit voltage out of range	▪ Switched ignition supply circuit - Voltage out of range	▪ Refer to the electrical circuit diagrams and check the switched ignition supply circuit to the control module for faults, including intermittent high resistance. Repair circuit, clear the DTCs and retest
U300A-62	Ignition Switch - Signal compare failure	▪ Mismatch between hardwired ignition voltage supply and control module internal measured voltage	▪ Check the security and integrity of the switched ignition supply circuit connectors. Rectify ▪ Refer to the electrical circuit diagrams and check the switched ignition supply circuit to the control module for faults, including intermittent high resistance. Repair circuit, clear the DTCs and retest
U300A-64	Ignition Switch - Signal plausibility failure	▪ Switched ignition supply circuit - Short circuit to ground, open circuit, high resistance	▪ Check the security and integrity of the switched ignition supply circuit connectors. Rectify ▪ Refer to the electrical circuit diagrams and check the switched ignition supply circuit to the control module for short circuit to ground, open circuit, high resistance. Repair circuit, clear the DTCs and retest